A four-chromatic orientation without a rainbow directed P_4

Alex Chan
Harvard University and National Bureau of Economic Research
achan@hbs.edu

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Abstract

Jensen and Toft ask whether every orientation of a four-chromatic graph has a directed path meeting each color class exactly once in some proper four-coloring. We give an acyclic orientation of a graph with 13 vertices and 36 edges for which the answer is no. The graph has a unique proper four-color partition, and an examination of the possible middle arcs shows that none of its directed four-vertex paths is rainbow. We give a self-contained finite proof. The chromatic-number and no-rainbow assertions, including the quantifier over all proper four-colorings, are also verified in Lean.

1 The question and the example

For an orientation D of a graph G, a directed path is *rainbow* with respect to a proper coloring c if its vertices have pairwise distinct colors. Jensen and Toft [2, Section 13.7, pp. 214–215] ask whether, for every four-chromatic G and every orientation D of G, there are a proper four-coloring c and a directed path meeting each of the four color classes exactly once. Thus the path has four vertices. The choice of c is part of the question; excluding a rainbow path for one prescribed coloring would not suffice. The question belongs to the line of problems discussed by Berge [1] and Meyniel [3].

Theorem 1. *There is a four-chromatic graph G on 13 vertices and an acyclic orientation D of G such that no directed path of D meets each color class exactly once in any proper four-coloring of G.*

Let $V(G) = \{0, 1, \dots, 12\}$. Table 1 specifies every arc of D; the underlying unordered pairs are precisely the edges of G. There are 36 arcs, no loops, and no pair of opposite arcs. The order

$$0, 3, 11, 4, 5, 7, 8, 9, 12, 1, 2, 6, 10$$

is a topological ordering, so D is acyclic. Write c_0 for the coloring with classes

$$\begin{aligned} c_0^{-1}(0) &= \{0, 1, 2, 3\}, \quad c_0^{-1}(1) = \{4, 5, 6\}, \\ c_0^{-1}(2) &= \{7, 8, 9\}, \qquad c_0^{-1}(3) = \{10, 11, 12\}. \end{aligned} \tag{1}$$

Every arc in Table 1 joins distinct classes, so c_0 is proper.

Table 1: Complete out-neighbor lists for the 36-edge orientation.

v	$N^+(v)$	v	$N^+(v)$
0	$4, 6, 8, 9, 12$	7	$1, 2, 6, 10, 12$
1	10	8	$2, 6$
2	$\varnothing$	9	$6, 10$
3	$6, 7, 10, 12$	10	$\varnothing$
4	$1, 2, 8, 10, 12$	11	$1, 2, 5, 6, 7, 8, 9$
5	$1, 9, 12$	12	$1, 2$
6	$\varnothing$		

Table 2: Exhaustive forcing cases. The notation $v : i$ means $c(v) = i$.

$(c(11), c(7))$	Forced assignments, in order	Outcome
$(1, 2)$	$12 : 3,\ 0 : 0,\ 1 : 0,\ 5 : 2,\ 6 : 3$	Vertex 9 sees colors $0, 1, 2, 3$ at $0, 11, 5, 6$.
$(1, 3)$	$6 : 0,\ 0 : 3,\ 9 : 2,\ 10 : 0,\ 1 : 2$	Vertex 12 sees colors $0, 1, 2, 3$ at $2, 4, 1, 0$.
$(3, 1)$	$6 : 0,\ 0 : 3,\ 12 : 2,\ 1 : 0,\ 3 : 3,\ 5 : 1,\ 9 : 2$	Vertex 10 sees colors $0, 1, 2, 3$ at $1, 4, 9, 3$.
$(3, 2)$	$1 : 0,\ 10 : 3,\ 12 : 3,\ 0 : 0,$ $6 : 1,\ 3 : 0,\ 9 : 2,\ 5 : 1$	The coloring c_0 in (1).

2 Uniqueness of the color partition

Lemma 1. *Every proper four-coloring of G has the classes in* (1)*, up to a permutation of the color names. Consequently $\chi(G) = 4$.*

Proof. Consider a proper coloring with colors $0, 1, 2, 3$. Since $2, 4, 8$ form a triangle, we may permute the color names so that their colors are $0, 1, 2$, respectively. Vertex 11 is adjacent to 2 and 8, so its color is 1 or 3. Vertex 7 is adjacent to 2 and 11. There are therefore exactly four possible ordered pairs $(c(11), c(7))$, listed in Table 2.

Read the assignments in each row from left to right. At each step the assigned vertex has already colored neighbors in the three other colors, so its displayed color is forced. In the first row, for example, the neighbors $2, 4, 7$ force $c(12) = 3$. Each of the first three rows ends with a vertex having neighbors in all four colors, as specified in the last column. These rows are impossible. The fourth row assigns all remaining vertices and gives precisely c_0, which is proper.

This proves uniqueness of the partition. A proper three-coloring could be viewed as a four-coloring with one unused color, whereas every class in (1) is nonempty. No proper three-coloring exists, and $\chi(G) = 4$. □

3 The directed paths

Lemma 2. *No directed path on four vertices in D is rainbow with respect to c_0.*

Proof. Write such a path as $x \to u \to v \to y$. Its middle arc $u \to v$ must have a predecessor of u and a successor of v. Exactly the nine arcs in Table 3 have this property. For each, the table lists all possible x and y and the resulting color words $c_0(x)c_0(u)c_0(v)c_0(y)$. The displayed predecessor and successor sets are disjoint from the middle vertices and from each other, so every listed choice

Table 3: Every possible middle arc of a directed four-vertex path.

$u \to v$	Possible x	Possible y	Color words	Paths
$4 \to 1$	0	10	0103	1
$4 \to 8$	0	2, 6	0120, 0121	2
$4 \to 12$	0	1, 2	0130	2
$5 \to 1$	11	10	3103	1
$5 \to 9$	11	6, 10	3121, 3123	2
$5 \to 12$	11	1, 2	3130	2
$7 \to 1$	3, 11	10	0203, 3203	2
$7 \to 12$	3, 11	1, 2	0230, 3230	4
$12 \to 1$	0, 3, 4, 5, 7	10	0303, 1303, 2303	5

is a path on four distinct vertices. Every color word has a repeated digit. The counts sum to 21, and no path is rainbow. □

Proof of Theorem 1. Lemma 1 gives $\chi(G) = 4$, and the displayed topological ordering proves acyclicity. Every proper four-coloring differs from c_0 only in its color names, which does not change whether a path is rainbow. Lemma 2 therefore excludes a rainbow directed four-vertex path for every proper four-coloring. □

A 37-edge variant. Adding $3 \to 4$ preserves the displayed topological ordering. Every proper four-coloring of the enlarged graph is proper for G, and c_0 remains proper. Thus the enlarged graph is four-chromatic with the same unique partition. Since 3 has no incoming arc, every new directed four-vertex path starts with $3 \to 4$. The five new paths have color words

$$0103, \quad 0120, \quad 0121, \quad 0130, \quad 0130.$$

None is rainbow, so the 37-edge orientation is another witness, with 26 directed four-vertex paths.

Formal verification. Lean 4.35.0 also verifies both finite constructions: looplessness, absence of opposite arcs, existence of a proper four-coloring, impossibility of a proper three-coloring, and the no-rainbow assertion for every proper four-coloring. The last assertion covers all directed walks of three arcs. The formal proof includes the color-class equalities in Lemma 1. Source and graph data are available in the accompanying repository.[1] The preceding argument is independent of this verification.

References

[1] C. Berge, Problem, in *Combinatorics 79, Part II*, M. Deza and I. G. Rosenberg, eds., Annals of Discrete Mathematics **9**, North-Holland, 1980, p. 297.

[2] T. R. Jensen and B. Toft, *Graph Coloring Problems*, Wiley, New York, 1995.

[3] H. Meyniel, About colorings, stability and paths in directed graphs, *Discrete Math.* **74** (1989), 149–150.

[1] https://github.com/alexgocardinal/jt137-rainbow-paths. The final certificates are `JT137.counterexample36` and `JT137.counterexample37`.